MILITARY SPECIFICATION

CAPACITORS, FIXED, FEED THROUGH, ELECTROMAGNETIC INTERFERENCE SUPPRESSION, (AC, DC, AND AC-DC), (HERMETICALLY SEALED IN METAL CASES),

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for AC, DC, and AC-DC, current carrying, feed-through capacitors, hermetically sealed in metal cases (see 6.4), for use primarily in suppression of electromagnetic interference.

1.2 Classification. Capacitors shall be classified by the dielectric specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-39028 - Capacitors, Packaging of.
MIL-C-45662 - Calibration System Requirements.
MIL-T-10727 - Plating, Tin.

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-220 - Method of Insertion-Loss Measurement.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MS35333 - Washer, Lock, Flat Internal.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

D92-57 - Method of Test for Flash and Fire Points by Cleveland Open Cup.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern. (See 6.1.)

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.2).

3.3 Materials. Materials shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Impregnant and filling compounds (when applicable, see 3.1). Compounds used in the impregnation and filling of capacitors shall be chemically inactive with respect to the capacitor element and the case. The compound either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor.

3.4 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Case. Each capacitor shall be enclosed in a hermetically-sealed metal case which will prevent leakage of the impregnant or filling compound, and will protect the capacitor element from moisture and mechanical damage under all test conditions specified herein. When specified (see 3.1), the case shall be plated in accordance with MIL-T-10727.

3.4.2 Finish. All exposed metal surfaces shall be suitably protected against corrosion by plating, or other means. The resultant finish shall form a good electrical conductor and shall be free from defects that may affect its protective value.

3.4.3 Threaded parts. All threaded parts shall be in accordance with Handbook H28 and shall be as specified (see 3.1). Aluminum nuts shall not be used.

3.4.3.1 Engagement of threaded parts. All threaded parts shall engage by at least two full threads.

3.5 Flashpoint of impregnant or filling compound. When measured as specified in 4.7.1, the flashpoint of the impregnant or filling compound shall be not lower than 165°C, for the temperature range specified (see 3.1).

3.6 Seal. When capacitors are tested as specified in 4.7.3, there shall be no evidence of leakage.

3.7 Terminal strength. When capacitors are tested as specified in 4.7.4, no part of the terminal shall loosen or rupture, and there shall be no other damage. Terminals shall exhibit no perceptible movement relative to the case, under the applied pull.

3.8 Barometric pressure (reduced). When tested as specified in 4.7.5, capacitors shall withstand the test voltage without visible evidence of external damage, flashover, or open- or short-circuiting.

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3.9 Dielectric withstanding voltage (not applicable to tantalum solid electrolyte). When capacitors are tested as specified in 4.7.6, there shall be no visible damage, flashover, or open- or short-circuiting.

3.10 Insulation resistance (when specified, see 3.1). When measured as specified in 4.7.7, the insulation resistance shall be not less than the value specified (see 3.1).

3.11 DC leakage (when specified, see 3.1). When measured as specified in 4.7.8, the dc leakage shall not exceed the value specified (see 3.1).

3.12 Capacitance. When measured as specified in 4.7.9, the capacitance shall be within specified tolerance of the value specified (see 3.1).

3.13 Dissipation factor. When measured as specified in 4.7.10, the dissipation factor shall not exceed the value specified (see 3.1).

3.14 Insertion loss. When measured as specified in 4.7.11 or 4.7.11.1, the insertion loss shall be not more than 6 decibels (dB) removed from the value shown in figure 1. Dips below the value shown on figure 1 (unless otherwise specified, see 3.1) for an ideal capacitor of the same nominal capacitance value are permissible when tested at frequencies from 0.050 megahertz (MHz) up to that frequency at which the insertion loss of the capacitor under test becomes 60 dB; however, at all higher frequencies up to maximum rated frequency, the insertion loss shall not fall below 60 dB. Deviations in the nature of dips in the curve will be permitted (such dips indicate a drop in insertion loss below that of an ideal capacitor) followed by a rise in insertion loss, as frequency is increased faster than the rise that is characteristic of an ideal capacitor.

3.15 Overload. When capacitors are tested as specified in 4.7.12, the insulation resistance shall be as specified (see 3.1) and the voltage drop shall not exceed the value specified (see 3.1).

3.16 DC resistance. When measured as specified in 4.7.13, the dc resistance shall not exceed the value specified (see 3.1).

3.17 Resistance to soldering heat. When capacitors are tested as specified in 4.7.14, there shall be no damage to the capacitors or the terminal insulators which will cause electrical failure of the capacitors, or which will cause hermetically-sealed capacitors to leak. Chipping of terminal insulators alone shall not be cause for failure unless the chipping extends to the outer periphery.

3.18 Salt spray (corrosion). When capacitors are tested as specified in 4.7.15, there shall be no harmful or extensive corrosion and at least 90 percent of any exposed metal surface of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test.

3.19 Thermal shock and immersion. When tested as specified in 4.7.16, capacitors shall meet the following requirements:

Dielectric withstanding voltage - -	As specified in 3.9.
Insulation resistance (when specified, see 3.1) - - - -	Not less than the value specified (see 3.1).
DC leakage (when specified, see 3.1) - - - - - - - - - - - -	Not less than the value specified (see 3.1).
Capacitance - - - - - - - - - - - - -	Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.9.
Dissipation factor - - - - - - - -	Shall not exceed the value specified (see 3.1).

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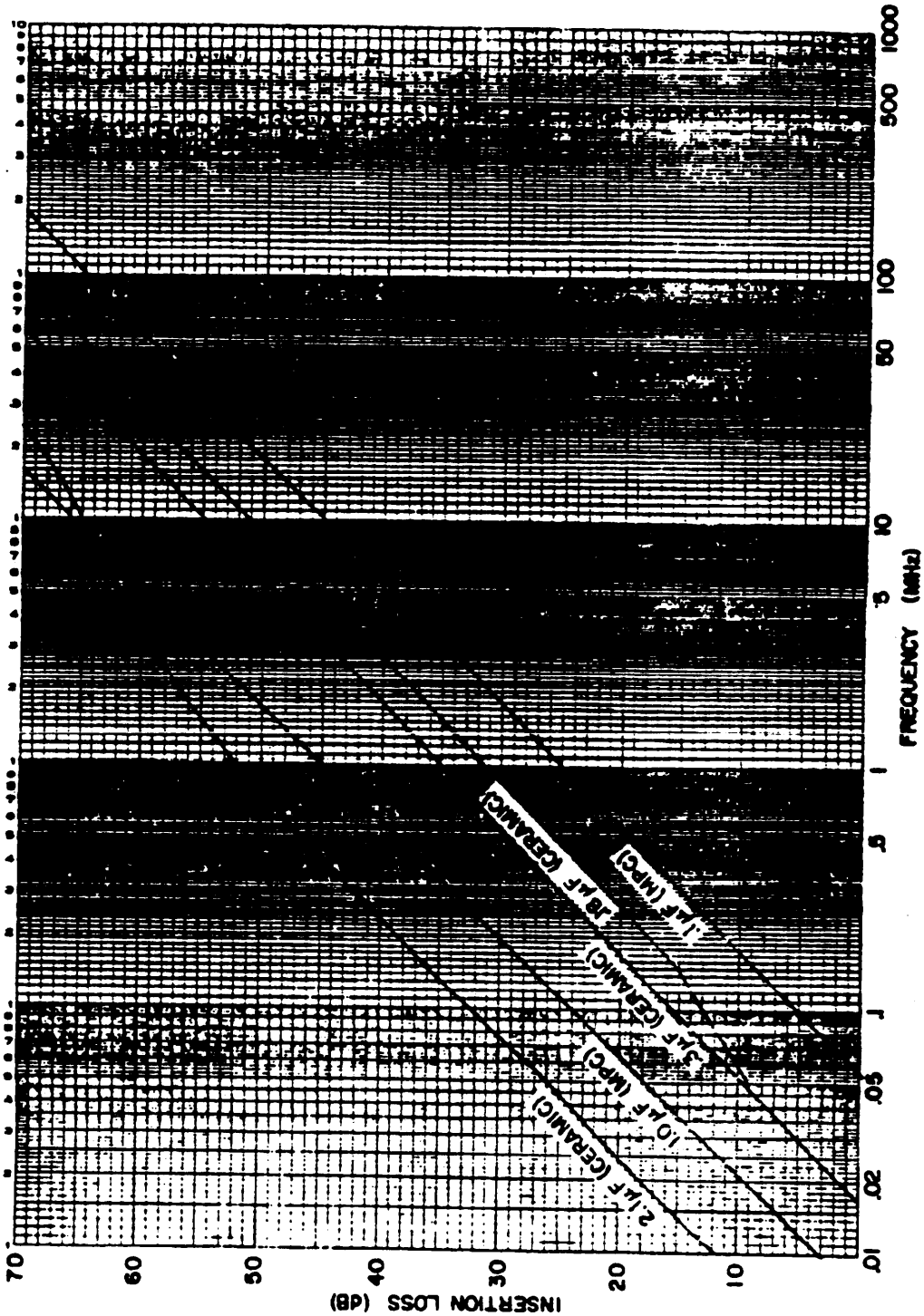


FIGURE 1. Insertion-loss versus frequency for an ideal capacitor.

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Insertion loss (abbreviated)	Change at any frequency not more than 3 dB from the initial value obtained when measured as specified in 4.7.11.1 except that for those frequencies at which the initial value is 60 dB or greater, the change may exceed 3 dB provided the insertion loss does not drop below 60 dB.
Visual examination	There shall be no harmful or extensive corrosion and at least 90 percent of any exposed metal surface of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test.

3.20 Shock (specified pulse). When capacitors are tested as specified in 4.7.17, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, open- or short-circuiting, and no mechanical damage to the capacitors. After completion of the shock test, the units shall be subjected to the abbreviated insertion loss test (see 4.7.11.1). The change at any frequency shall be not more than 3 dB from the initial value obtained when measured as specified (see 3.1), except for those frequencies at which the initial value is 60 dB or greater, which may exceed 3 dB provided the insertion loss does not drop below 60 dB.

3.21 Vibration, high frequency. When capacitors are tested as specified in 4.7.18, there shall be no intermittent contacts of 0.5 ms or greater duration, open- or short-circuiting and no mechanical damage to the capacitors.

3.22 Moisture resistance. When tested as specified in 4.7.19, capacitors shall meet the following requirements:

Insulation resistance (when specified, see 3.1)	Not less than the value specified (see 3.1).
DC leakage (when specified, see 3.1)	Shall not exceed the value specified
Capacitance	Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.9.
Dissipation factor	Shall not exceed 1.0 percent.
Visual examination	There shall be no harmful or extensive corrosion and at least 90 percent of any exposed metal surface of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test.

3.23 Solderability. When capacitors are tested as specified in 4.7.20, the dipped portion of the terminals shall conform to the solid-wire termination criteria of method 208 of MIL-STD-202.

3.24 Resistance to solvents. When capacitors are tested as specified in 4.7.21, there shall be no evidence of mechanical damage, and the marking shall remain legible.

3.25 Capacitance change at reduced temperature. When capacitors are tested as specified in 4.7.22, there shall be no evidence of breakdown, flashover, or fracture. The capacitance change at -55° ±3°C shall not exceed 15 percent of the value measured at 25°C.

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3.26 Life. When tested as specified in 4.7.23.1 or 4.7.23.2, capacitors shall meet the following requirements:

Insulation resistance (when specified, see 3.1)	Not less than the value specified (see 3.1).
DC leakage (when specified, see 3.1)	Shall not exceed the value specified (see 3.1).
Capacitance	Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.9.
Dissipation factor	Shall not exceed the value specified (see 3.1).
Insertion loss (abbreviated)	Unless otherwise specified (see 3.1), change at any frequency not more than 3 dB from the initial value obtained when measured as specified in 4.7.11.1, except that for those frequencies at which the initial value is 60 dB or greater, the change may exceed 3 dB provided the insertion loss does not drop below 60 dB.
Visual examination	There shall be no mechanical failure or leakage of impregnant or filling compound.

3.27 Marking. Capacitors shall be marked in accordance with method I of MIL-STD-1285, and shall include the military part number, source code, date code, and voltage and current ratings. The marking shall be placed on the exposed cylindrical surface of the case and shall remain legible after all environmental tests. Other markings which in any way interfere with, obscure, or confuse those specified herein are prohibited.

3.28 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

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4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample units produced with equipment and procedures normally used in production.

TABLE I. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of defectives allowed 1/
Group I				
Flashpoint of impregnant or filling compound	3.5	4.7.1	---	
Visual and mechanical examination (internal): Materials, design, construction, and workmanship	3.1, 3.3, 3.4, and 3.28	4.7.2	2	0
Visual and mechanical examination (external): Physical dimensions, marking 2/, and workmanship	3.1, 3.3, 3.4, and 3.29	4.7.2	47 3/	1
Seal	3.6	4.7.3		
Terminal strength	3.7	4.7.4		
Barometric pressure (reduced)	3.8	4.7.5		
Dielectric withstanding voltage (not applicable to tantalum solid electrolyte)	3.9	4.7.6		
Insulation resistance (when specified, see 3.1)	3.10	4.7.7		
DC leakage (when specified, see 3.1)	3.11	4.7.8		
Capacitance	3.12	4.7.9		
Dissipation factor	3.13	4.7.10		
Insertion loss	3.14	4.7.11		
Overload	3.15	4.7.12		
DC resistance	3.16	4.7.13		
Group II				
Resistance to soldering heat	3.17	4.7.14		1
Salt spray (corrosion)	3.18	4.7.15	6	
Thermal shock and immersion	3.19	4.7.16		
Group III				
Shock (specified pulse)	3.20	4.7.17		
Vibration, high frequency	3.21	4.7.18	6	
Moisture resistance	3.22	4.7.19		
Group IV				
Solderability	3.23	4.7.20	4	1
Resistance to solvents	3.24	4.7.21	4	
Group V				
Capacitance change at reduced temperature	3.25	4.7.22	24	1
Life at +85°C	3.26	4.7.23.1	12	
Life at +125°C (when applicable, see 3.1)	3.26	4.7.23.2	12	

1/ A specimen having one or more defects shall be considered as a single defective.
2/ Marking shall be considered as a defect only if it becomes illegible as a result of any of the tests of the specification.
3/ One additional sample unit is included in each 47 sample units to permit substitution for failure allowed in Group I.

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4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in table I and the appendix to this specification.

4.4.2 Test routine. The specimens shall be subjected to the qualification inspection specified in table I in the order shown. Two specimens in a sample shall be subjected to visual and mechanical examination (internal). The remaining specimens shall be subjected to the subsequent examinations and tests in group I. The specimens shall then be divided into the remaining groups specified in table I, and subjected to the tests for their particular group.

4.4.3 Failures. Failures in excess of those allowed in table I shall be cause for refusal to grant qualification approval.

4.5 Retention of qualification. To retain qualification, the supplier shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) A summary of the results of tests performed for periodic inspection (group B), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 6-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each dielectric to testing in accordance with the qualification inspection requirements.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all capacitors of the same dielectric or electrolyte material, and current and voltage rating produced under essentially the same conditions and presented for inspection at the same time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II in the order shown.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	Acceptable quality level (percent defective)	
			Major	Minor
Subgroup 1				
Seal	3.6	4.7.3	1.0	---
Dielectric withstanding voltage (not applicable to tantalum solid electrolyte)	3.9	4.7.6		
Insulation resistance (when specified, see 3.1)	3.10	4.7.7		
DC leakage (when specified, see 3.1)	3.11	4.7.8		
Capacitance	3.12	4.7.9		
Dissipation factor	3.13	4.7.10		
Insertion loss	3.14	4.7.11		
DC resistance	3.16	4.7.13		
Subgroup 2				
Visual and mechanical examination (external): Materials, physical dimensions, design and construction (other than physical dimensions), marking 1/, and workmanship	3.1, 3.3, 3.4, 3.28, and 3.29	4.7.2	1.0	4.0

1/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect markings. Any subsequent electrical defects shall not be used as a basis for determining marking defects.

4.6.2 Periodic inspection. Periodic inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.5), delivery of products which have passed group A shall not be delayed pending the results of these periodic inspections.

4.6.2.1 Group B inspection. Group B inspection shall consist of the tests specified in table III, in the order shown, and shall be performed on sample units selected from lots that have passed group A inspection. Test data obtained shall be reviewed as a part of the complete retention of qualification.

4.6.2.1.1 Sampling plan. Sample units shall be representative of capacitors produced.

4.6.2.1.2 Subgroups 1, 2, 3, and 4. Twenty-four sample units shall be taken from production every six months and subjected to the applicable tests for their particular subgroup. Allowable failures shall be as specified in table III.

4.6.2.1.3 Subgroup 5. Twelve sample units shall be taken from production every six months and subjected to the applicable tests for their particular subgroup. Allowable failures shall be as specified in table III.

4.6.2.1.4 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

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TABLE III. Group B inspection.

Test	Requirement paragraph	Method paragraph	Number of sample units to be inspected	Number of defectives allowed
Subgroup 1				
Terminal strength	3.7	4.7.4		
Barometric pressure (reduced)	3.8	4.7.5	6	1
Insulation resistance (when specified, see 3.1) (at the high ambient temperature)	3.10	4.7.7		
Insertion loss	3.14	4.7.11		
Overload	3.15	4.7.12		
Subgroup 2				
Resistance to soldering heat	3.17	4.7.14	6	1
Salt spray (corrosion)	3.18	4.7.15		
Thermal shock and immersion	3.19	4.7.16		
Subgroup 3				
Shock (specified pulse)	3.20	4.7.17	6	1
Vibration, high frequency	3.21	4.7.18		
Moisture resistance	3.22	4.7.19		
Subgroup 4				
Solderability	3.23	4.7.20	3	1
Resistance to solvents	3.24	4.7.21	3	
Subgroup 5				
Capacitance change at reduced temperature	3.25	4.7.22	12	1
Life at +85°C	3.26	4.7.23.1	6	
Life at +125°C (when applicable, see 3.1)	3.26	4.7.23.2	6	

4.6.2.1.5 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-C-39028.

4.7 Methods of examination and test.

4.7.1 Flashpoint of impregnant or filling compound (see 3.5). This test is applicable only to liquid impregnants and filling compounds and to solid impregnants having a melting point of less than 135°C. The flashpoint of the impregnant or filling compound shall be measured as specified in Publication D92-57, except that fire point and precision do not apply. "Impregnant or filling compound" shall be substituted for the word "oil" throughout the test method.

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4.7.2 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.27, and 3.28).

4.7.3 Seal (see 3.6). Capacitors shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply:

(a) Test-condition letter - As specified (see 3.1).
(b) Measurement after test - Not applicable.

4.7.4 Terminal strength (see 3.7). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - As specified (see 3.1).
(b) Method of holding - Normal mounting means.
(c) Applied force - As specified (see 3.1).

4.7.5 Barometric pressure (reduced) (see 3.8). Capacitors shall be tested in accordance with method 105 of MIL-STD-202, The following details shall apply:

(a) Method of mounting - By normal mounting means.
(b) Test-condition letter - C.
(c) Test during subjection to reduced pressure - Test voltage of 125 percent of rated voltage shall be applied. A suitable means shall be used to detect momentary or permanent breakdown.
(d) Examination after test - Capacitors shall be examined for evidence of damage.

4.7.6 Dielectric withstanding voltage (not applicable to tantalum solid electrolyte (see 3.9). Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude and nature of test voltage - 200 percent of rated dc voltage, or a dc potential equivalent to 300 percent of rated ac voltage (see 3.1), whichever is applicable.
(b) Duration of application of test voltage - 1 minute, minimum.
(c) Points of application of test voltage - Between the case and either terminal.
(d) Limiting value of surge current - 50 milliamperes.
(e) Examination after test - After the test, capacitors shall be visually examined for external evidence of damage, flashover, or breakdown.

4.7.7 Insulation resistance (when specified, see 3.1) (see 3.10). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details and exception shall apply:

(a) Test-condition letter - As specified (see 3.1).
(b) Maximum charging current - 50 milliamperes.
(c) Measurements shall be made between the case and all line (not grounded) terminals of the same circuit connected together.

4.7.8 DC leakage (when specifed, see 3.1)(see 3.11). DC leakage shall be measured using the dc rated voltage ±2 percent at the applicable test temperature (see 3.1), after a maximum electrification period of 5 minutes. A 1,000-ohm resistor shall be placed in series with the capacitor to limit the charging current. A steady source of power, such as a regulated power supply, shall be used. Measurement accuracy shall be within ±2 percent or 0.02 microampere (μA), whichever is greater. Units shall be stabilized at the rated temperature for 30 minutes prior to applications of voltage.

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4.7.9 Capacitance (see 3.12). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details shall apply:

(a) Test frequency - As specified (see 3.1).
(b) Limit of accuracy - Within ±2 percent.
(c) Magnitude of polarizing voltage - When required (see 3.1).

4.7.10 Dissipation factor (see 3.13). The dissipation factor of each capacitor shall be measured at a root-mean-square (rms) voltage not greater than 20 percent of the rated voltage, ac or dc, whichever is applicable (see 3.1). Measurement shall be made at a frequency of 1 kHz ±100 Hz. Measurement accuracy shall be within ±0.1 percent absolute. For tantalum solid electrolyte capacitors, the dissipation factor shall be measured at a frequency of 120 ±5 Hz by means of a polarized capacitance bridge. The inherent accuracy of the measurement shall be ±(2 percent actual value +0.001).

4.7.11 Insertion loss (see 3.14). Capacitors shall be tested in accordance with MIL-STD-220. Measurements shall be taken at a sufficient number of frequencies to permit the construction of a smooth accurate curve of insertion loss versus frequency over a frequency range of .050 to 1,000 MHz inclusive. Suggested frequencies are as follows: .050, 0.10, 0.15, 0.5, 1, 2, 4, 8, 14, 20, 40, 80, 120, 200, 300, 400, and 1,000 MHz.

4.7.11.1 Insertion loss (abbreviated). Capacitors shall be tested as specified in 4.7.11, except that measurements shall be made only at the frequencies specified (see 3.1).

4.7.12 Overload (see 3.15). Capacitors shall be suspended by their terminals in free air at an ambient temperature of 125°C minimum. Rated current shall then be applied for 15 minutes. After the capacitors have reached thermal stability, and while still energized, the voltage drop shall not exceed the value specified (see 3.1). Capacitors shall be visually examined for evidence of physical damage.

4.7.13 DC resistance (see 3.16). Capacitors shall be tested in accordance with method 303 of MIL-STD-202. The maximum allowable measurement error shall be ±5 percent.

4.7.14 Resistance to soldering heat (see 3.17). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

(a) Depth of immersion in molten solder - Within .062 ±.031 inch of the seal or case.
(b) Test-condition letter - B.
(c) Cooling time prior to measurement after test - 1 hour.
(d) Measurement after test - Capacitance and dc resistance shall be measured as specified in 4.7.9 and 4.7.13, respectively.

4.7.15 Salt spray (corrosion) (see 3.18). Capacitors shall be tested in accordance with method 101, test-condition letter A of MIL-STD-202. After the test, capacitors shall be visually examined for corrosion and obliteration of marking.

4.7.16 Thermal shock and immersion (see 3.19).

4.7.16.1 Thermal shock. Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - A (unless otherwise specified, see 3.1), except that in step 3, the high ambient test temperature shall be $125^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$C.
(b) Measurements before and after cycling - Not applicable.

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4.7.16.2 Immersion. Within 4 to 24 hours after completion of temperature cycling, capacitors shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

(a) Test-condition letter - A (unless otherwise specified, see 3.1).
(b) Measurements shall be made within 2 hours.
(c) Measurements after final cycle - Unless otherwise specified (see 3.1), dielectric withstanding voltage, insulation resistance, capacitance, dissipation factor, and insertion loss (abbreviated) shall be measured as specified in 4.7.6, 4.7.7, 4.7.9, 4.7.10, and 4.7.11.1, respectively.

4.7.17 Shock (specified pulse) (see 3.20). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting - Securely fastened by normal mounting means. When applicable, leads and connecting wires shall be supported to prevent damage to the capacitor.
(b) Test-condition letter - I.
(c) Electrical load during shock - During the test, a potential of 125 percent of rated voltage (unless otherwise specified, see 3.1) shall be applied to the capacitor. The charging current shall be limited to 50 milliamperes.
(d) Measurements during shock - During the test, a cathode ray oscilloscope or other comparable means capable of detecting intermittent contacts of 0.5 ms duration or greater shall be used as an indicating device in determining any electrical failures.
(e) Examination after test - Capacitors shall be visually examined for evidence of arcing, breakdown, or mechanical damage.

4.7.18 Vibration, high frequency (see 3.21). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting - Securely fastened by normal mounting means. When applicable, leads and connecting wires shall be so supported to exclude spurious resonances.
(b) Electrical load conditions - During the test, a potential of 125 percent of rated voltage (unless otherwise specified, see 3.1) shall be applied between the terminals and case.
(c) Test-condition letter - D (unless otherwise specified, see 3.1).
(d) Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cyclindrical axis.
(e) Measurements during vibration - During the last half hour of vibration in each direction, a signal of 1 ±0.2 kHz at a level of 1 ±0.2 volts shall be placed across the capacitor and measured with a suitable device to determine open- or short-circuiting or intermittents of 0.5 ms or greater duration.
(f) Examination after test - Capacitors shall be visually examined for evidence of mechanical damage.

4.7.19 Moisture resistance (see 3.22). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Initial measurements - Not applicable.
(b) Polarization voltage - 50 vdc or rated voltage (see 3.1), whichever is less, unless otherwise specified, shall be applied to either terminal and the case of 50 percent of the capacitors. No potential shall be applied to the remaining 50 percent of the capacitors.

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(c) Final measurements - Unless otherwise specified (see 3.1), upon completion of step 6 of the final cycle, insulation resistance, capacitance, and dissipation factor shall be measured as specified in 4.7.7, 4.7.9, and 4.7.10, respectively, at a relative humidity of 50 ±5 percent.
(d) Visual examination - Capacitors shall be examined for corrosion, mechanical damage, and obliteration of marking.

4.7.20 Solderability (see 3.23). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following detail shall apply:

(a) The number of terminations of each capacitor to be tested - 2.

4.7.21 Resistance to solvents (see 3.24). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) Portion of specimen to be brushed - That portion on which marking is present.
(b) Number of specimens to be tested - As specified in applicable inspection tables.
(c) Permissible extent of damage - As specified in 3.24.

4.7.22 Capacitance change at reduced temperature (see 3.25). Capacitor shall be placed in a chamber and maintained at -55° ±3°C with rated voltage and current applied for a period of 48 ±4 hours. Before the capacitor is removed from the chamber, the capacitance shall be measured as specified in 4.7.9, except temperature shall be at -55° ±3°C. The capacitance change shall not exceed the value measured in 4.7.9 by more than the value specified (see 3.1).

4.7.23 Life (see 3.26).

4.7.23.1 Life at +85°C. Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

(a) Distance between specimens - 1.5 inch, minimum.
(b) Test temperature - 85° $^{+4°}_{-0°}$C.
(c) Operating conditions - Voltage as specified (see 3.1) shall be applied between terminal and case. Capacitors shall have rated current flowing through the feed-through terminals.
(d) Test-condition letter - As specified (see 3.1).
(e) Measurements after exposure - Capacitors shall be returned to the inspection conditions specified in 4.3, and shall be visually examined for leakage of impregnant or filling compound and deformation of case. Insulation resistance or dc leakage (as specified, see 3.1), capacitance, dissipation factor, and insertion loss (abbreviated) shall then be measured as specified in 4.7.7, 4.7.8, 4.7.9, 4.7.10, and 4.7.11.1, respectively.

4.7.23.2 Life at +125°C (when applicable, see 3.1). Capacitors shall be tested as specified in 4.7.23.1, with the following exceptions:

(a) Test temperature shall be +125° $^{+4°}_{-0°}$C at the voltage specified (see 3.1).
(b) Test-condition letter - As specified (see 3.1).

5. PREPARATION FOR DELIVERY

5.1 Capacitors shall be prepared for delivery in accordance with MIL-C-39028.

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6. NOTES

6.1 Ordering data.

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet (see 3.1).
(c) Levels of preservation and packaging and packing, and applicable marking (see section 5).

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Air Force; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

6.2.1 SD-6, "Provisions Governing Qualification," is issued for the information of applicants requesting qualification of products. Copies of this publication may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.3 Insertion loss for ideal capacitors. The insertion loss of ideal capacitors for capacitance values from 0.001 to 100 µF, inclusive, and frequency range from 0.01 to 1,000 megahertz, inclusive, may be determined from figure 2.

6.4 Definition of hermetic sealing. For the purpose of this specification, a hermetically-sealed capacitor is one in which the capacitance element is contained within a sealed enclosure of ceramic, glass or metal, or combinations thereof, where sealing is accomplished by material fusion, welding, brazing or soldering.

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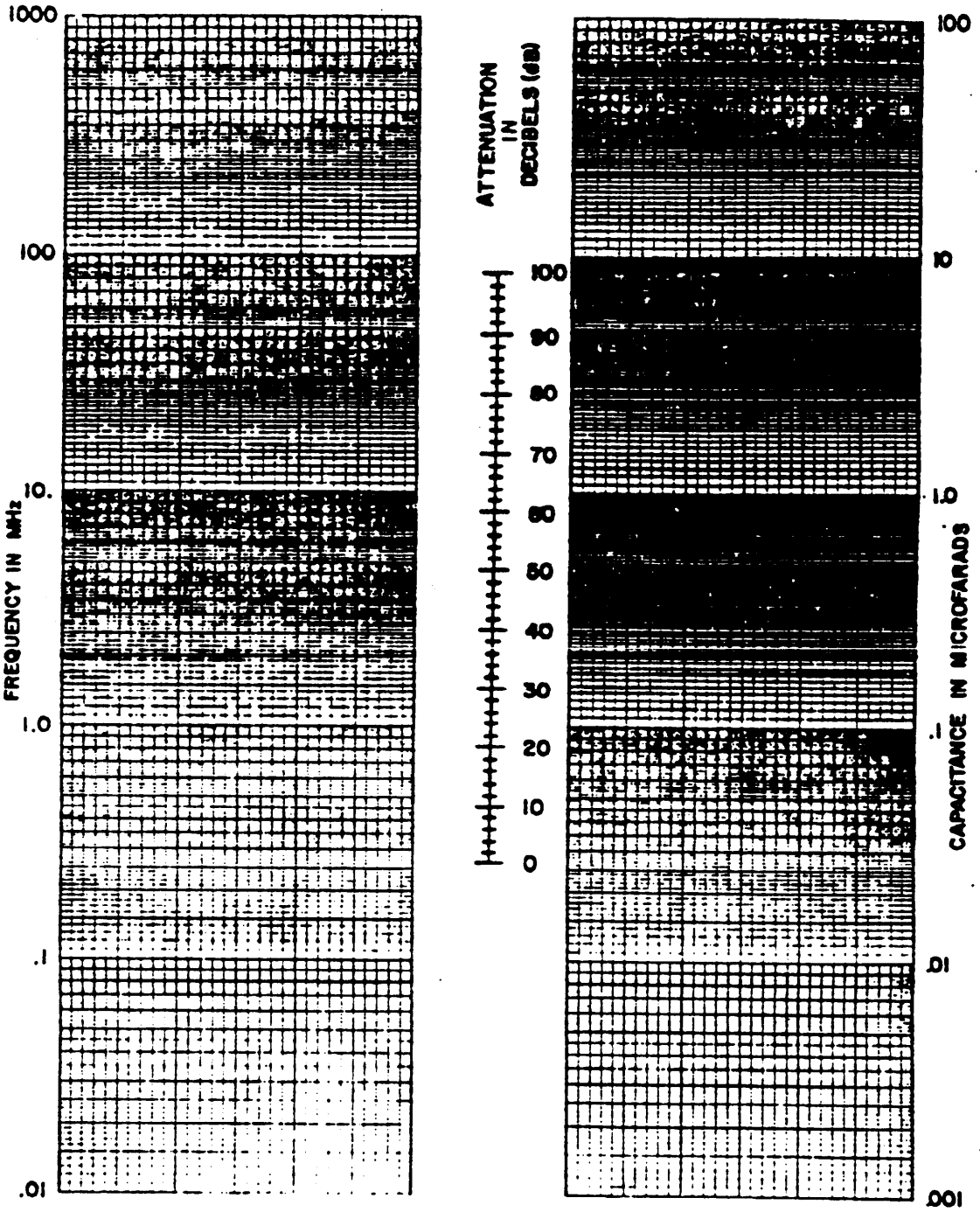


FIGURE 2. Insertion-loss chart for ideal capacitors.

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 Sample.

20.1.1 Single-type submission. A sample consisting of 47 specimens of the type for which qualification is sought shall be submitted (see 30.1). Each sample shall be accompanied with the following information:

(a) Attenuation curves in accordance with MIL-STD-220.
(b) Either a certification of the flashpoint of the impregnant or filling compound as determined in accordance with 4.7.1 or a minimum of 200 cubic centimeters of the impregnant or filling compound used in the capacitors.

20.1.2 Combined-type submission. A maximum of two types in each style for which qualification is sought, consisting of 24 specimens of the highest watt-second rating or volt-ampere reactive, as applicable in the highest voltage rating and 24 specimens of the lowest capacitance value in any voltage rating, shall be submitted. Each sample shall be accompanied by the information required in 20.1.1(a) and 20.1.1(b), above.

20.2 Description of items. The manufacturer shall submit a detailed description of the materials and constructional features of the capacitors being submitted for inspection, including information on whether they are liquid-filled or liquid-impregnated; the type and quantity of the impregnant or filling compound; the type, thickness, and number of layers of capacitor tissue and electrode material; and the material, thickness, and applied finish of the case.

30. EXTENT OF QUALIFICATION

30.1 Single-type submission. Qualification shall be restricted to the type submitted.

30.2 Combined-type submission. Qualification of the types submitted shall be extended to types whose watt-second and voltage ratings are not greater than the highest watt-second and voltage ratings submitted, and whose capacitance values are not less than the lowest capacitance value submitted. In addition, submission and qualification of a type or types in one automatically carries with it qualification of the lower, provided the units are of the same style, material, impregnant or filling compound and construction.

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